

The Spacetime Origin Axiom (SOA): A Geometric Framework for Price-Time Squaring

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Abstract

Market trajectories are often viewed as stochastic processes; however, through the lens of **Ho's Spacetime Origin Axiom (SOA)**, they can be understood as predictable vectors defined by an initial velocity. By calculating a fixed **Rate of Change (ROC)** from an All-Time Low (ATL) to a determined spacetime target, we can construct a "Center Line" that acts as the gravitational axis for price action. This paper outlines the methodology for projecting these trajectories and utilizing geometric envelopes to define market volatility.

I. Methodology: The SOA Workflow

The SOA algorithm operates on the premise that a stock's long-term trend is "locked" once its origin and primary target are identified. The process follows five rigorous steps:

1. **Primary Target Determination:** Utilizing the SOA algorithm to identify the terminal price and time for the primary wave (e.g., a target of 380 for META within a 10-year horizon).
2. **Origin Identification:** Locating the All-Time Low (ATL) price and date to serve as the spacetime anchor.
3. **Vector Calculation:** Determining the constant **Rate of Change (ROC)** or "Rate of Rally" required to traverse the distance between the ATL and the SOA target.
4. **Forward Projection:** Using the ROC and ATL date to project the "Center Line"—a 1x1 trajectory of price-time equilibrium.
5. **Anniversary Verification:** Observing price action at the "Anniversary Date" or its immediate vicinity. According to Gann's price-time squaring requirement, the price must intersect or "touch" the trend line on these dates to maintain the integrity of the cycle.

II. The Center Line and Geometric Envelopes

The **Center Line** (the ROC projection) is the immutable core of the model. While the price may fluctuate, the ROC remains fixed once the SOA target is set. To account for market "noise" and high-volatility rallies, we establish **Upper and Lower Envelopes**.

- **The Upper Envelope:** Acts as "Hard Resistance." During a strong bullish rally, the price may overextend, but it will inevitably correct back toward the Center Line as it approaches the anniversary date.
- **The Lower Envelope:** Acts as "Structural Support." In periods of correction, this line provides the floor where the price stabilizes before returning to its mean trajectory.

These envelopes are not arbitrary. They are derived from Gann's divisions of a circle (360), using intervals such as **45, 90, or 180 units**. While the envelopes can be adjusted based on the specific historical volatility of the asset, the **Center Line remains absolute**, or can be recalibrated with the anniversary data points as trading time progresses.

III. Empirical Demonstrations: TEM and META

To illustrate the SOA in practice, we examine the long-term charts of **TEM** and **META**.

- **TEM Analysis:** By anchoring at the ATL on 06/25/2024 and applying the SOA target of 370 within the 10-year window, the resulting ROC creates a diagonal that price has respected with remarkable symmetry. Note how deviations toward the upper envelope are met with corrections that "square" exactly at the anniversary of the origin. The upper and lower envelope lines are 45 points vertically from the center trend line.



Fig. 1, Notice the price is near the center trend line around the first anniversary date, 06/25/2025, from the 06/25/2024 ATL point.

- **META Analysis:** Similarly, for META, the target of 380 within the 10-year window provides a roadmap where the price-time intersections act as pivots for major trend continuations. The upper and lower envelop lines are 45 points from the center trend line.



Fig. 2, Earlier years from its IPO in 2012.



Fig. 3, Later years between 2017 and 2019, showing the price movements between and upper and lower trend lines.

IV. Conclusion

The SOA algorithm provides a roadmap for the "Value of Time." Unlike traditional indicators that lag behind price, the SOA-Gann synthesis allows the researcher to stand at the origin and see the destination. By respecting the Center Line and monitoring the geometric envelopes, an investor can distinguish between a healthy trend and a terminal overextension.

Note to Readers

The logic of the SOA indicator is based on the proprietary Ho-Gann Doctrine. While the specific scripts for these calculations are withheld, the mathematical principles of ROC and price-time squaring are accessible to any practitioner of technical geometry who wishes to implement this framework within their own analytical suite.